Q1:(Please write your answers in a table(below) only, otherwise, your answers will be neglected)

1. Keyboard and traditional monitors are examples of
[✓ A. Simplex devices] B. Duplex devices C. Half Duplex devices

2. Protocols are a set of rules to govern
[✓ A. Communication] B. Standards C. Metropolitan communication

3. The parameter that refers to the uneven delay of data packets in the delivery is
[✓ A. Jitter] B. Timelessness C. Accuracy

4. In mesh topology, the devices are connected via
A. Multipoint link [✓ B. Point to point link] C. No Link

5. Hybrid network includes a combination of
A. Nodes B. Devices [✓ C. topologies]

6. Delivering data to the correct destination is
[✓ A. Delivery] B. Accuracy C. Timelessness

7. The transmission impairment that refers to a signal with high energy in a very short time is:
A. Induced Noise B. Crosstalk [✓ C. Impulse Noise]

8. The Internet is a:
A. Collection of WANS [✓ B. Network of networks] C. Collection of LANS

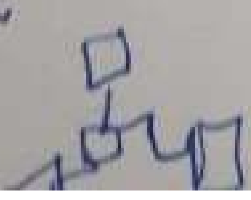
9. The mode that is like a two-way street with traffic flowing in both directions simultaneously is
A. Simplex [✓ B. Full Duplex] C. Half Duplex

10. Two computers connected by an Ethernet hub are of
[✓ A. LAN topology] B. MAN topology C. WAN topology

11. The following topology has no central controlling unit:
[✓ A. mesh topology] B. star topology C. LAN topology

12. The impractical dedicated network is: شبكة مخصصة غير عملية
[✓ A. mesh topology] B. star topology C. LAN topology

غالية



13. The traditional mail system is:
[✓ A. half duplex] B. full duplex C. simplex

14. +1dB represents an increase in power of:
[✓ A. 2.6%] B. 26% C. 260%

$+1\,dB = 10\log_{10} \frac{P_{in}}{P_{out}}$

$10 =$

$0.1 = \log_{10} \frac{P_a}{P_{in}}$

$P_{in}\, 10^{0.1} = P_{out}$

15. The word Tele means:
[✓ A. Far] B. Near C. Apart

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	A	B	C	A	C	B	B	A
Question	11	12	13	14	15					
Answer	A	A	A	A	A					A

+13

Q2: The characteristics of certain audio amplifier are such that it gives a voltage amplification of 10 at 100Hz, 30 at 3kHz and 60 at 10kHz. Taking the amplification at 3kHz as the reference level, calculate the loss or gain in decibels at the other two frequencies. (5 marks)

3KHZ – with – 100Hz

input $V=30$ out $V=10$

$$\text{gain dB} = 20\log_{10}\frac{V_o}{V_{in}} = 20\log_{10}\frac{100}{3\times10^3} \quad , V_{in} > V_{out}\ (1)\ (2)\ \text{loss}$$

$$\boxed{-29.54\,dB = \text{loss in dB}}$$

3KHZ with 10kHz

$V_{out} > V_{in}$ (2) (1) Gain

Gain

$$\text{gain dB} = 20\log_{10}\frac{(10k)}{(3k)} = 10.45\,dB$$

Gain in dB

لسل سؤال ستقوم بتحليله
لدينا 3 ترددات
وله فولتية معينة
100Hz V=10
3KHZ N=30
10KHZ, V=60

نزيد حسابه Gain
و loss باعتبار
3KHz
هو مرجعنا
عندما نزيد الحساب
100Hz
و 10KHz و

Q3: a. Suppose that a digitized TV picture is to be transmitted from a source that uses a matrix of 480 × 500 picture elements (pixels), where each pixel can take on one of 32 intensity values. Assume that 30 pictures are sent per second. (This digital source is roughly equivalent to broadcast TV standards that have been adopted.) Find the source rate R (bps).

MHz

b. Assume that the TV picture is to be transmitted over a channel with 4.5-MHz bandwidth and a 35-dB signal-to-noise ratio. Find the capacity of the channel (bps).

bit Rata $= 480 * 500 * \underset{\text{pixel}}{30} = \frac{\text{pixel}}{s}$

$= 7.2 * 10^6$

$\boxed{= 7.2 \text{ ~~mb~~ ~~d~~ Mpbs}}$

(b) first

SNR dB = $10 \log_{10} \text{SNR}$

$\frac{35}{10} = \frac{10}{10} \log_{10} \text{SNR}$

$\log_{10} \text{SNR} = 3.5$

$\text{SNR} = 10^{3.5}$

≈ 3162.2

$= \underline{\underline{3162}}$

Shannon

$C = B * \log_2 (1 + \text{SNR})$

$B = 4.5 \text{MHZ} = 4.5 * 10^6$

SNR = 35dB

$= 4.5 M \log_2 (1 + 3162)$

$\boxed{= 52.32 \text{ MbPs}}$ Mega bit per second

GOOD LUCK